

LDscnR — LD-aware outlier regions

Finding the loci underlying adaptation by using linkage disequilibrium as evidence rather than as a nuisance.

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github.com/genomeinsights/LDscnR

THE PREMISE

Genome scans confuse signal with structure. Small samples, complex demography and relatedness all produce peaks that look like selection. **LD tells them apart.** Selection locally inflates linkage disequilibrium — a causal variant drags a block of correlated markers with it — so a true signal appears as a *cluster* of mutually correlated markers, while a false positive arising from relatedness sits isolated, with low LD to its neighbours.

“LD-supported” and “clustered” are therefore not two separate criteria but the same footprint, seen once per marker and once per region. Everything below follows from that.

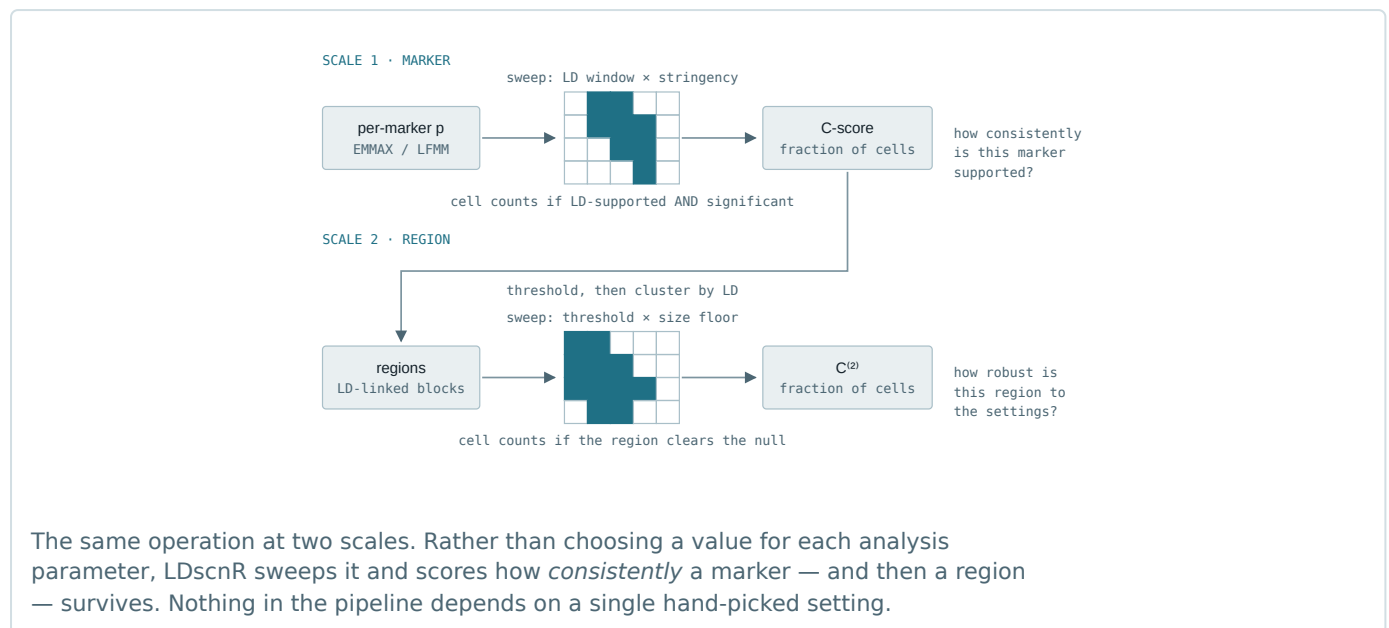
THE PIPELINE

00 Detectability Can this genome be scanned at all? A truth-free check that signal is separable from background, so structure-dominated datasets are flagged rather than over-interpreted.	01 Kinship Build the relatedness matrix from LD- <i>independent</i> markers, so it cannot absorb the very regions under selection. Stringency set by the genomic inflation factor.
02 Association scan EMMAX, LFMM, or any test giving one <i>p</i> -value per marker. The method consumes the vector and is agnostic to its source.	03 Marker C-score Sweep the LD window and the filtering stringency. C = the fraction of settings in which a marker is <i>both</i> LD-supported <i>and</i> significant.
04 Outlier regions Cluster surviving markers by <i>linkage</i> , not by physical distance — so an inversion stays one unit while an adjacent looser cluster separates.	05 Structure-aware null Surrogate phenotypes that keep the population structure but break the link to the environment. Each region gets a location-matched, FDR-controlled <i>p</i> -value.
06 Second-tier C-score Sweep the region threshold and the size floor. $C^{(2)}$ = the fraction of settings in which a locus stays significant — a robustness ranking.	07 Report Ranked loci with two numbers: is it real (region-level FDR), and how robust is it to the analyst's parameter choices.

WHAT IS DIFFERENT ABOUT IT

Filtering that costs nothing Keeping only LD-supported markers before correcting for multiple testing is <i>independent filtering</i> : it raises power by shrinking the tested set without breaking error control, because local LD is independent of the association statistic under the null.	Regions, not SNPs Markers are grouped by linkage rather than by physical distance, and the joining threshold self-calibrates to each chromosome's LD decay. Multiple testing is then reported over tens of regions, not hundreds of thousands of SNPs.	No cherry-picking Every analyst-chosen parameter is integrated over instead of fixed, and the reported score says how much of that parameter space supports the locus. A result that holds in only one corner of the grid is visible as such.
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THE IDEA IN ONE PICTURE



HOW IT DIFFERS FROM EXISTING APPROACHES

Haplotype and LD-pattern sweep scans ^{H12} ·

OmegaPlus

These read elevated LD as direct evidence of selection. LDscnR does not — elevated LD also arises from inversions, recombination-rate variation, demography and structure. Local LD is used only as an *independent* measure of genomic coherence for prioritising association signals; it is never interpreted as a test of selection and never folded into the evidence.

LD-based complexity reduction ^{the LDna lineage}

LDscnR descends from LD-network clustering, and inherits its unsolved problem. Li *et al.* (2018) cut tests by replacing each cluster with its principal components — cheaper, but the component is weaker than the strongest SNP in the region. Fang *et al.* (2021) kept the SNP-level statistics and swept the clustering parameters across 36 settings, then reported the setting that recovered the most regions. LDscnR takes the same sweep and **integrates over it rather than selecting from it**: choosing the best-performing cell reintroduces, at the parameter level, the multiple-testing problem being controlled within it.

Window-based regional aggregation ^{WZA}

Fixed physical windows assume adjacent variants carry one evolutionary signal; recombination means they often do not, so aggregation dilutes highly localised signal. LDscnR takes regions from the topology of the local LD network — overlapping but distinct signals in the same interval stay distinct — and preserves the original SNP-level statistics instead of summarising them.

How significance is established

Permutation is the acknowledged gold standard for multiple-testing correction, because it accounts for the correlation structure of the genome directly. Under population structure the standard version is invalid — reshuffling phenotypes destroys the among-individual correlation the model assumes — and the correct substitute is already specified in the literature: draw surrogate phenotypes from $MVN(0, \sigma_g^2 A + \sigma_e^2 I)$ and rerun the scan on each. What has limited it is cost, and cost scales with the number of tests. Li *et al.* bought affordability by dimension reduction, at the price of signal. **LDscnR buys it by carrying the per-SNP burden earlier**, so the null need only resolve a handful of regions per surrogate — 50–200 surrogates rather than tens of thousands — with the SNP-level statistics kept.

STATUS

Validated on forward-in-time, individual-based simulations (NEMO) with known causal variants, and applied to a three-spined stickleback marine–freshwater dataset (117 individuals, 35 populations, 47–75°N) where single-SNP scans resolve nothing. There it recovers the known architecture — the chromosome-1 inversion, the *Eda* region, and peaks on chromosomes 5, 7, 17 and 20 — with the regions independently corroborated against a published set of recurrent marine–freshwater adaptation peaks from separate cohorts and a different assembly.

Implemented as an R package: github.com/genomeinsights/LDscnR. Manuscript in preparation — comments welcome at petri@genomeinsights.fi.